

## FOURTH ORDER MAGNETIC MOMENT OF THE ELECTRON

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**Abstract:** A more complete account is given of an estimation, by rigorous upper and lower bounds, of the fourth order magnetic moment of the electron which has already been published by the author. The method is illustrated for one typical term. Then the results of a more elaborate estimation are given. In the terminology used by Karplus and Kroll, the term symbolized by the diagram IIc is found to satisfy

$$\mu(\text{IIc}) = (-0.60 \pm 0.11) \frac{\alpha^2}{\pi^2} - \frac{1}{2} \frac{\alpha^2}{\pi^2} \log \frac{\lambda^2}{m^2},$$

in contradiction with the value

$$-3.18 \frac{\alpha^2}{\pi^2} - \frac{1}{2} \frac{\alpha^2}{\pi^2} \log \frac{\lambda^2}{m^2}$$

given by the previous authors. It has been found that the other terms are not in disagreement with Karplus and Kroll values. If one accepts them, then the total fourth order moment amounts to  $\mu^{(4)} = (-0.39 \pm 0.11)(\alpha^2/\pi^2)$ . One notes finally that this new value increases the Lamb shift by  $(0.82 \pm 0.03)$  Mc/s and brings the fine structure constant to  $1/\alpha = 137.0384$ .

### 1. Introduction

The new experimental determination of the electron  $g$ -factor done by Franken and Liebes <sup>1)</sup> has been the real starting point of a doubt concerning the exactness of the theoretical calculations performed some years ago by R. Karplus and N. Kroll <sup>2)</sup> †. However, such a feeling was already expressed by W. E. Lamb in 1953 <sup>3)</sup> in connection with a slight disagreement between the observed and the calculated  $2^2S_{\frac{1}{2}} - 2^2P_{\frac{1}{2}}$  shift in hydrogen and deuterium, which clearly exceeded both the experimental and the theoretical uncertainties. Immediately a check was undertaken by the author. To be as fast as possible, the term by term check has been performed by using a method of rigorous and lower upper bounds. To obtain the safest results the inequalities for the lower bounds were derived for the integrals involving many auxiliary variables. This, of course, gave the simplest integrals, consequently the safest, but did not provide very narrow bounds. Nevertheless, whereas for diagrams I, IIa, IIc, and IIe (in the KK terminology) the bounds agreed with the values given in their paper, the inequality  $\mu(\text{IIc}) \geq -1.55 (\alpha^2/\pi^2)$  (infra-red term disregarded) was obtained for the diagram IIc. A slackening of this lower bound was allowed to make obvious the contradiction with  $-3.18 (\alpha^2/\pi^2)$  quoted by KK. This result has been published <sup>4)</sup>. Then, deriving for this term more involved inequalities on the

† Denoted in the following by KK.

many-variable integrals, better bounds have been found to yield, without the infra-red term,  $-0.49 (\alpha^2/\pi^2) \geq \mu(\text{IIc}) \geq -0.71 (\alpha^2/\pi^2)$ . In the present paper, section 2 will give, for the diagram IIc only, the detailed auxiliary-variable integrals. Then, in section 3, a typical term will be examined in detail as an example of the inequality technique for the many-variable integrals, illustrating its simplicity. Finally, in section 4, numerical results will be quoted concerning the diagram IIc, the total moment, and other observable electro-dynamical quantities such as the Lamb-shift, the moment of the mu-meson and the fine structure constant, taking into account the new value of the theoretical electron moment.

## 2. Auxiliary-Variable Integrals

No mention will be made here of the diagrams I, IIa, IId, and IIe, these not being in disagreement with the bounds calculated. It is of course not claimed that these four terms are correct, but simply that they are not in contradiction with the checking bounds†. For IIc, details are given. First of all, this term is expressed as follows:

$$\begin{aligned} \text{IIc} = & \frac{ie\alpha^2}{4\pi^4\hbar c} \frac{1}{(2\pi)^4} \int d^4 p_1 d^4 p_2 \int d^4 k \bar{\psi}(p_1) k^{-2} \int_0^1 u du \int_0^1 dv \\ & \times \gamma_\nu (i\gamma p_1 - i\gamma k + m)^{-1} \gamma_\mu (i\gamma p_2 - i\gamma k + m)^{-1} [A_1 + A_2 + A_3 + B + C \\ & + D + E + F + H + J] \psi(p_2), \end{aligned} \quad (2.1)$$

with the definitions

$$\begin{aligned} A_1 &= -m(1-u^2)(i\gamma p_2 - i\gamma k + m) \gamma_\nu A'^{-2}(p_2 - k, p_2), \\ A_2 &= -i(2p_{2\nu} - k_\nu)(1-u)(1-uv)(i\gamma p_2 - i\gamma k + m) A'^{-2}, \\ A_3 &= -ik_\nu(1-u+2uv)(1-uv)(i\gamma p_2 - i\gamma k + m) A'^{-2}, \\ B &= -(1-u) \gamma_\nu(1-uv)[(p_2 - k)^2 + m^2] A'^{-2}, \\ C &= ik_\nu \mu u(1+u)(1-2v) A'^{-2}, \\ D &= k^2 \gamma_\nu(1-u+u^2 v(1-v)) A'^{-2}, \\ E &= -k_\alpha \sigma_{\nu\alpha} \mu u(1-u) A'^{-2}, \\ F &= \int_0^1 dz \gamma_\nu \frac{A'^2 - A_0^2}{A_0^2 + (A'^2 - A_0^2)z}, \\ H &= -2m^2 \left(1-u-\frac{u^2}{2}\right) \gamma_\nu A_0^{-2}, \\ J &= 2m^2 \left(1-u-\frac{u^2}{2}\right) \gamma_\nu A'^{-2}, \\ A'^2(p_2 - k, p_2) &= u^2 m^2 + k^2 u^2 v(1-v) + u(1-u)v[(p_2 - k)^2 + m^2], \\ A_0^2 &= u^2 m^2. \end{aligned} \quad (2.2)$$

† Note added in proof: A further analytic evaluation performed in the meantime by the author<sup>7)</sup> shows that in fact  $\mu(\text{I})$  does not check with the value of Karplus and Kroll. The discrepancy amounts to  $+0.031 \alpha^2/\pi^2$ .

Furthermore, after combination of the denominators and integration over  $k$ , the contributions to the fourth order magnetic moment from the terms  $A_1 - J$  of (22) are given, in units of  $\alpha^2/\pi^2$ , by

$$\begin{aligned}
 \mu(A_1) &= (1-u^2)(1-w\beta\alpha^{-1})T, \\
 \mu(A_2) &= -(1-u)(1-t)w(1-u+uv)T, \\
 \mu(A_3) &= 0, \\
 \mu(B) &= (1-u)(1-u+uv)w\beta\alpha^{-1}(1-w\beta\alpha^{-1})T, \\
 \mu(C) &= 0, \\
 \mu(D) &= -(1-u+u^2v(1-v))w(1-\beta\alpha^{-1})[1-w(1-\beta\alpha^{-1})]T', \\
 \mu(E) &= -u(1-u)(1-w\beta\alpha^{-1})T \\
 &\quad + 2u(1-u)x(1-w(1-\beta\alpha^{-1}))[1-x(1-w(1-\beta\alpha^{-1}))]T_j + u(u-1)T'_j, \\
 \mu(F) &= -u^2v(1-v)[w(1-\beta\alpha^{-1})(1-w(1-\beta\alpha^{-1}))]T'_z \\
 &\quad - u(1-u)(1-v)w\beta\alpha^{-1}(1-w\beta\alpha^{-1})T_z, \\
 \mu(H) &= -\frac{5}{4} - \frac{1}{2} \log \frac{\lambda^2}{m^2}, \\
 \mu(J) &= 2 \left( 1-u - \frac{u^2}{2} \right) x(1-w(1-\beta\alpha^{-1}))[1-x(1-w(1-\beta\alpha^{-1}))]T_j,
 \end{aligned} \tag{2.3}$$

with the abbreviations (integrations over  $u, v, t, w, z$  and  $x$  from 0 to 1 understood)

$$\begin{aligned}
 \beta &= u(1-v)[1-u+uv(1-t)], \\
 \alpha &= u(1-v)(1-u+uv), \\
 T &= (1-u+uv)\{ut(1-u+uv)+w(1-v)[1-u+uv(1-t)]^2\}^{-1}, \\
 T' &= w(1-u+uv)\{utw(1-u+uv)+(1-v)[1-u+uv(1-wt)]^2\}^{-1}, \\
 T_z &= (1-u+uv)\{ut(1-u+uv)+zw(1-v)[1-u+uv(1-t)]^2\}^{-1}, \\
 T'_z &= w(1-u+uv)\{utw(1-u+uv)+z(1-v)[1-u+uv(1-wt)]^2\}^{-1}, \\
 T_j &= w(1-v)(1-u+uv)^3 \\
 &\quad \times \{utw(1-u+uv)+x(1-v)[1-u+uv(1-wt)]^2\}^{-2}, \\
 T'_j &= wx(1-u+uv)\{utw(1-u+uv)+x(1-v)[1-u+uv(1-wt)]^2\}^{-1}.
 \end{aligned} \tag{2.4}$$

As shown in (2.3), the infra-red cut-off  $\lambda^2$  appears only in  $H$ . In the following, it will be dropped, being understood that it has been taken into account to compensate the same term with opposite sign occurring in IIId. Once (2.3) and (2.4) are obtained by means of well-known Dirac algebra tricks and particular combinations of the denominators by means of Feyn-

man variables<sup>†</sup>, the next problem is to get estimates for the different contributions. First, special attention has been given to the lower bound in order to put in evidence, in the simplest way, the fact that  $\mu(\text{IIc})$  could not reach the value  $-3.18 (\alpha^2/\pi^2)$  Bohr magnetons. The next section will show how the inequality technique for the original many-auxiliary-variable integrals has been applied. An illustration of it will be performed on  $\mu(A_1)$ .

### 3. Inequality on a Four-Variable Integral

According to (2.3) and (2.4), the contribution of the  $A_1$  term in (2.1) to the fourth order magnetic moment is given, in units of  $\alpha^2/\pi^2$ , by

$$\mu(A_1) = \int_0^1 \int_0^1 \int_0^1 \int_0^1 (1-u^2) [1-w(1-u+uv(1-t))(1-u+uv)^{-1}] \times (1-u+uv) \times \{ut(1-u+uv) + w(1-v)[1-u+uv(1-t)]^2\}^{-1} du dv dw dt. \quad (3.1)$$

It is first noticed that  $\mu(A_1)$  is positive definite. Every factor is indeed positive definite, as is trivially seen for  $(1-u^2)$ ; for  $(1-w(1-u+uv(1-t)))(1-u+uv)^{-1}$  the same is true since the coefficient of  $w$  is always contained between 0 and 1. Finally, the last term in the integrand is obviously positive as  $1-u+uv$  is. To get a lower bound, we then look for an inequality of the form

$$\mu(A_1) \geq L. \quad (3.2)$$

One increases the denominator of the last term, thus decreasing the corresponding integral, by replacing  $[1-u+uv(1-t)]^2$  by  $1-u+uv$  and then dividing numerator and denominator by  $1-u+uv$ . Moreover,  $[1-u+uv(1-t)](1-u+uv)^{-1}$  being always positive and less than 1, one decreases the integral by replacing this factor by 1 in the second term, i.e. writing  $(1-w)$  instead of  $1-w[1-u+uv(1-t)](1-u+uv)^{-1}$ . Then one gets

$$\mu(A_1) \geq \int_0^1 \int_0^1 \int_0^1 \int_0^1 (1-u^2)(1-w)[ut+uv]^{-1} du dv dw dt, \quad (3.3)$$

the integral over  $v$  being unchanged when  $1-v$  is replaced by  $v$ . Displaying the first two factors:

$$(1-u^2)(1-w) = 1-w-u^2+wu^2, \quad (3.4)$$

four integrals, corresponding to each of the separate terms in (3.4) are to be computed.

The term proportional to 1,  $\int_0^1 \dots \int_0^1 [ut+uv]^{-1} du dv dw dt$ , is first integrated over  $t$ , giving

<sup>†</sup> The way of combining denominators used here has been found to be very suitable to get easy bounds owing to the positive definiteness of special combinations of Feynman variables. Different ways may exist, having the same properties, but many others have been experimented and found more complicated than the present ones.

$$\int_0^1 \dots \int_0^1 \frac{1}{u} [\log(u+vw) - \log vw] du dv dw. \quad (3.5)$$

Then, an integration over  $w$  leads to

$$\int_0^1 \int_0^1 \left\{ \left( \frac{1}{u} + \frac{1}{v} \right) \log(u+v) - \frac{1}{v} \log u - \frac{1}{u} \log v \right\} du dv. \quad (3.6)$$

The expression (3.6) is integrated first over  $u$  or  $v$ , depending on the terms, to give

$$\begin{aligned} & \int_0^1 \frac{1}{u} \log(1+u) du + \int_0^1 \frac{1}{v} \log(1+v) dv + \int_0^1 \log(1+u) du \\ & + \int_0^1 \log(1+v) dv - \int_0^1 \log u du - \int_0^1 \log v dv = 4.42. \end{aligned} \quad (3.7)$$

For the term proportional to  $w$ , after an integration over  $t$ , one is led to compute

$$\int_0^1 \int_0^1 \int_0^1 \frac{w}{u} \log \frac{u+vw}{vw} du dv dw. \quad (3.8)$$

The integral is first performed over  $v$ , then over  $w$ , then over  $u$ . It is expressed in terms of elementary functions and  $\int_0^1 u^{-1} \log(1+u) du$ . It yields

$$\int_0^1 \int_0^1 \int_0^1 du dv dw \frac{w}{u} \log \frac{u+vw}{vw} = 1.56. \quad (3.9)$$

For the term proportional to  $u^2$ , one integrates in the order  $t, w, u, v$  to get, always in terms of the same functions,

$$\int_0^1 \dots \int_0^1 u^2 [ut+vw]^{-1} dv du dw dt = 0.92. \quad (3.10)$$

The term proportional to  $wu^2$  is integrated in the order  $t, v, u, w$  to yield

$$\int_0^1 \dots \int_0^1 wu^2 [ut+vw]^{-1} dw du dv dt = 0.36. \quad (3.11)$$

The value  $L$  of (3.2) is now obtained by collecting, according to (3.4), (3.7), (3.9), (3.10) and (3.11) with the result<sup>†</sup>

$$\mu(A_1) \geq 2.30. \quad (3.12)$$

Similar techniques can be applied as simply to the other terms in (2.3), taken separately or in special combinations, grouping for instance the first and second term of  $E$  with  $A_1, A_2, B$  and  $J$  respectively. It will be observed that the most cumbersome bound was that of the  $J$  kind. Here the technique outlined for  $A_1$  was used in the interval  $0 \leq u \leq 0.1$ . For  $0.1 \leq u \leq 1$ , the first mean value theorem was applied.

The results for these elementary lower bounds are

<sup>†</sup> Note added in proof: The analytic value for  $\mu(A_1)$  has been found to be 2.60.

$$\begin{aligned}
\mu(A_1 + A_2 + B + \text{first term of } E) &\geq +1.45 \frac{\alpha^2}{\pi^2} \\
\mu(D) &\geq -0.11 \frac{\alpha^2}{\pi^2} \\
\mu(\text{third term of } E) &\geq -0.23 \frac{\alpha^2}{\pi^2} \\
\mu(F) &\geq -0.15 \frac{\alpha^2}{\pi^2} \\
\mu(H) &= -1.25 \frac{\alpha^2}{\pi^2} \\
\mu(J + \text{second term of } E) &\geq -1.26 \frac{\alpha^2}{\pi^2}
\end{aligned} \tag{3.13}$$

and

$$\mu(\text{IIc}) \geq -1.55 \frac{\alpha^2}{\pi^2}.$$

After this result has been obtained, it becomes clear that the value  $-3.18 (\alpha^2/\pi^2)$ , reported in KK, is erroneous and it is then worthwhile to compute more elaborate bounds in order to determine more precisely the value of IIc and, consequently, that of the total fourth order moment. These bounds will be quoted in the following section together with the new values of some physical quantities affected by the change of the magnetic moment.

#### 4. Numerical Values

Deriving the auxiliary integrals, sometimes with a combination of the denominators different from those used for the elementary bounds, and then deriving more involved inequalities for these, one gets much better bounds. However, the calculations are much more involved too, without presenting special interest. In this way,  $\mu(\text{IIc})$  has been shown to satisfy <sup>†</sup>

$$-0.49 \frac{\alpha^2}{\pi^2} \geq \mu(\text{IIc}) \geq -0.71 \frac{\alpha^2}{\pi^2}. \tag{4.1}$$

Accepting now the numerical results reported by Karplus and Kroll for  $\mu(\text{I})$ ,  $\mu(\text{IIa})$ ,  $\mu(\text{IId})$  and  $\mu(\text{IIe})$ , the total fourth order moment would turn out to satisfy <sup>††</sup>

<sup>†</sup> The upper bound in (4.1) had already been obtained when ref. <sup>4)</sup> was published. The improvement of the present work on the latter has been a tightening of the lower bound which was previously kept intentionally very broad so as to provide the simplest proof that  $\mu(\text{IIc})$  was inexact in ref. <sup>3)</sup>.

<sup>††</sup> This is in good agreement with a calculation done by C. Sommerfield <sup>5)</sup> who finds, by a quite different method,  $\mu^{(4)} = -0.33(\alpha^2/\pi^2)$ .

$$-0.28 \frac{\alpha^2}{\pi^2} \geq \mu^{(4)} \geq -0.50 \frac{\alpha^2}{\pi^2}. \quad (4.2)$$

This is to be compared with the recent experimental value<sup>1)</sup>

$$\mu_{\text{exp}}^{(4)} = (+0.7 \pm 2.0) \frac{\alpha^2}{\pi^2}$$

expressed in the same units.

Taking for  $\mu^{(4)}$  the average value  $(-0.39 \pm 0.11) \alpha^2/\pi^2$  increases the Lamb shifts for hydrogen and deuterium to

$$(H) \quad 2^2S_{\frac{1}{2}} - 2^2P_{\frac{1}{2}} = (1057.92 \pm 0.21) \text{ Mc/s} \quad (4.3)$$

$$(D) \quad 2^2S_{\frac{1}{2}} - 2^2P_{\frac{1}{2}} = (1059.26 \pm 0.21) \text{ Mc/s}, \quad (4.4)$$

respectively, in excellent agreement with observation.

The electromagnetic fourth order magnetic moment of the  $\mu$ -meson taking into account the vacuum polarization by electrons<sup>5)</sup>, amounts to

$$\mu'^{(4)} = (+0.69 \pm 0.11) \frac{\alpha^2}{\pi^2} \frac{e\hbar}{2M_{\mu}c}, \quad (4.5)$$

$M_{\mu}$  being the mass of the  $\mu$ -meson.

Finally, the fine structure constant is slightly modified; it becomes

$$1/\alpha = 137.0384 \quad (4.6)$$

compared to the previous value 137.0365.

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